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USB_PD_R3_2 V1.1 2024-10_Chapter 10 Power Rules.pdf

20 pages (732 KB)

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Total Changes

482

Text only comparison

Content

326 Replacements

81 Insertions

75 Deletions

Styling and Annotations

0 Styling

0 Annotations

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10 Power Rules

10.1 Introduction

The flexibility of power provision on *USB Type-C®* is expected to lead to power adapter re-use and the increasingly widespread provision of USB power outlets in domestic and public places and in transport of all kinds. Environmental considerations could result in unbundled power adapters. Rules are needed to avoid incompatibility between the *Sources* and the *Sinks* they are used to power, in order to avoid user confusion and to meet user expectations. This section specifies a set of rules that *Sources* and *Sinks* **shall** follow. These rules provide a simple and consistent user experience.

The *PDP Rating* is a manufacturer declared value placed on packaging to help the user understand the capabilities of a *Charger* or the size of *Charger* required to power their device. For *PDP* values of 10W and above the *PDP* **shall** be declared as an integer number of Watts. For *PDP* values less than 10W, the *PDP* **shall** be declared in increments of 0.5W.

The *Source Power Rules* define a *PDP* to provide a simple way to tell the user about the capabilities of their power adapter or device. *PDP Rating* is akin to the wattage rating of a light bulb - bigger numbers mean more capability.

The *Sink Power Rules* define a *PDP* to provide a simple way to tell the user which *Sources* will provide adequate power for their *Sink*.

10.2 Source Power Rules

The *Source Power Rules* defined in this section include both **Normative** and **Optional** rules. For all of the defined rules, the capabilities a *Source* exposes are based on the **Port Maximum PDP**, or if power constrained, the **Port Present PDP** of the *Port*.

For a *Guaranteed Capability Port*, the *Source* **shall** always include in every *Source_Capabilities* or *EPR_Source_Capabilities* Message sent to a *Sink* all the (A)PDOs that are defined by the **Normative** (and **Optional** when implemented) rules based on the *Port's* **Port Maximum PDP** and *Mode* of operation (i.e., *SPR Mode* or *EPR Mode*).

For a *Managed Capability Port*, except before the *First Explicit Contract* or before the *Explicit Contract* after the **Port Present PDP** changes on a *Shared Capacity Charger Port*, the *Source* **shall** always include in every *Source_Capabilities* or *EPR_Source_Capabilities* Message sent to a *Sink* all the (A)PDOs that are defined by the **Normative** (and **Optional** when implemented) rules based on the *Port's* **Port Present PDP** and *Mode* of operation (i.e., *SPR Mode* or *EPR Mode*). After the *First Explicit Contract*, this requirement assures that the attached *Sink* will always know what voltages (or voltage modes) are presently available from the *Source*.

In order to meet the expectations of the user, the Maximum Current/Power in the *Source Capabilities* PDO or APDO for *Sources* with a *PDP Rating* of x Watts **shall** be as follows:

- Maximum current for **Normative** and **Optional** *Fixed Supply/Variable Supply* PDOs **shall** be either RoundUp(x/voltage) or RoundDown(x/voltage) to the nearest 10mA.
- Maximum current for *SPR Programmable Power Supply* APDOs **shall** be as defined in [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#).

Note: When the Constant Power bit is set in the APDO, the programmable power supply's output current is as defined in [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#) however the programmable power supply will limit its output current so that the product of its actual output voltage times the output current does not exceed the *PDP*.

- If a 9V Prog, 15V Prog or 20V Prog Programmable Power Supply APDO is *Advertised* when not required by [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#), then the maximum current **shall** be RoundDown (x/Prog Voltage) to the nearest 50mA. When the PPS Power Limited bit is clear the *Source* **shall** provide this current at **Maximum Voltage**.
- Maximum power for **Optional** *Battery Supply* PDOs **shall** be $\leq x$.

10.2.1 Source Power Rule Considerations

The *Source Power Rules* are designed to:

- Ensure the *PDP Rating* (*PDP*) of an adapter specified in watts explicitly defines the voltages and currents at each voltage the adapter supports.
- Ensure that adapters with a large *PDP Ratings* are always capable of providing the power to devices designed for use with adapters with a smaller *PDP Rating*.
- Enable an ecosystem of adapters that are inter-operable with the devices in the ecosystem.

The considerations that lead to the *Source Power Rules* are based are summarized in [Table 10.1, "Considerations for Sources"](#).

Table 10.1 Considerations for Sources

Considerations	Rationale	Consequence
Simple to identify capability	A user going into an electronics retailer knows what they need	Cannot have a complex identification scheme
Higher power Sources are a superset of smaller ones	Bigger is always better in user's eyes – don't want a degradation in performance	Higher power Sources do everything smaller ones do
Unambiguous Source definitions	Sources with the same power rating but different VI combinations might not inter-operate	To avoid user confusion, any given power rating has a single definition
A range of power ratings	Users and companies will want freedom to pick appropriate Source ratings	Fixed profiles at specific power levels don't provide adequate flexibility, e.g., profiles as defined in previous versions of PD.
5V@3A USB Type-C Source is defined by [USB Type-C 2.4]	5V@3A USB Type-C Source is considered	All > 15W adapters must support 5V@3A or superset consideration is violated
Maximize 3A cable utilization	3A cables will be ubiquitous	Increase to maximum voltage (20V) before increasing current beyond 3A
Optimize voltage rail count	More rails are a higher burden for Sources, particularly in terms of testing	5V is a basic USB requirement. 48V provides the maximum capability.
Some Sources are not able to provide significant power	Some small Battery-operated Sources e.g., mobile devices, are able to provide more power directly from their Battery than from a regulated 5V supply	In addition to the minimal 5V Advertisements are able to Advertise more power from their Battery
Some Sources share power between multiple Ports (Hubs and multi-Port Chargers)	Hubs and multi-port Chargers have to be supported	See Section 10.3, "Sink Power Rules"

10.2.2 Normative Voltages and Currents

The voltages and currents an *SPR Source* with a *PDP Rating* of x Watts **shall** support are as defined in [Table 10.2](#), "[SPR Normative Voltages and Minimum Currents](#)".

Table 10.2 **SPR Normative Voltages and Minimum Currents**

Port Maximum PDP Rating (W)	5V Fixed	9V Fixed	15V Fixed	20V Fixed	SPR AVS
$0.5 \leq x \leq 15$	$(PDP/5)A^3$	-	-	-	-
$15 < x \leq 27$	$3A^2$	$(PDP/9)A^3$	-	❌❌❌❌	-
$27 < x \leq 45$	$3A^2$	$3A^2$	$(PDP/15)A^3$	-	(9V – 15V): • (15V Fixed Supply Max Current) A
$45 < x \leq 60$	$3A^2$	$3A^2$	$3A^2$	$(PDP/20)A^3$	(9V – 15V): • (15V Fixed Supply Max Current) A ⁴ (15V – 20V): • (20V Fixed Supply Max Current) A
$60 < x \leq 100$	$3A^2$	$3A^2$	$3A^2$	$(PDP/20)A^{1,3}$	(9V – 15V): • (15V Fixed Supply Max Current) A ^{4,5} (15V – 20V): • (20V Fixed Supply Max Current) A ^{1,5}
1)	Requires a 5A cable.				
2)	The <i>Fixed Supply</i> PDOs Maximum Current field shall <i>Advertise</i> at least 3A, but may <i>Advertise</i> up to RoundUp (PDP/voltage) to the nearest 10mA. Requires a 5A cable if over 3A is <i>Advertised</i> .				
3)	The <i>Fixed Supply</i> PDOs Maximum Current field shall <i>Advertise</i> either RoundDown (PDP/voltage) or RoundUp (PDP/voltage) to the nearest 10mA.				
4)	<i>SPR AVS</i> current for this voltage range is the maximum current as <i>Advertised</i> by the 15V <i>Fixed Supply</i> PDO. This current can be higher than 3A (refer to Note 2). Requires a 5A cable if over 3A is <i>Advertised</i> .				
5)	The <i>Sink</i> is allowed to request up to the 20V <i>Fixed Supply</i> Max Current when the requested voltage is 15.0V.				

SPR *Managed Capability Ports* when power constrained are defined to offer **Valid (A)PDOs** based on the port's **Port Maximum PDP** (as per [Table 10.2, "SPR Normative Voltages and Minimum Currents"](#)) at lower **Port Present PDP** (as per [Table 10.3, "SPR Source Capabilities When Port Present PDP is less than Port Maximum PDP"](#)) because these voltages would otherwise be available if the *Managed Capability Port* power hadn't been constrained.

Table 10.3 SPR Source Capabilities When Port Present PDP is less than Port Maximum PDP

Port Present PDP (W)	5V Fixed	9V Fixed	15V Fixed	20V Fixed	SPR AVS with Max Voltage of 15V or 20V per Table 10.2 ⁶
$5 < x \leq 15$	$(PDP/5)A^3$	$(PDP/9)A^{3,7,8}$	$(PDP/15)A^{3,7,8}$	$(PDP/20)A^{3,7,8}$	(9V – 15V): (15V Fixed Supply Max Current) $A^{4,6,8}$ (15V – 20V): (20V Fixed Supply Max Current) $A^{6,8}$
$15 < x \leq 27$	$3A^2$	$(PDP/9)A^3$	$(PDP/15)A^{3,7}$	$(PDP/20)A^{3,7}$	(9V – 15V): (15V Fixed Supply Max Current) A^4 (15V – 20V): (20V Fixed Supply Max Current) A^6
$27 < x \leq 45$	$3A^2$	$3A^2$	$(PDP/15)A^3$		
$45 < x \leq 60$	$3A^2$	$3A^2$	$3A^2$	$(PDP/20)A^3$	(9V – 15V): (15V Fixed Supply Max Current) A^4 (15V – 20V): (20V Fixed Supply Max Current) A
$60 < x \leq 100$	$3A^2$	$3A^2$	$3A^2$	$(PDP/20)A^{1,3}$	(9V – 15V): (15V Fixed Supply Max Current) $A^{4,5}$ (15V – 20V): (20V Fixed Supply Max Current) $A^{4,5}$
1) Requires a 5A cable. 2) The <i>Fixed Supply</i> PDOs Maximum Current field Shall Advertise at least 3A, but May Advertise up to RoundUp $(PDP/voltage)$ to the nearest 10mA. Requires a 5A cable if over 3A is <i>Advertised</i> . 3) The <i>Fixed Supply</i> PDOs Maximum Current field Shall Advertise either RoundDown $(PDP/voltage)$ or RoundUp $(PDP/voltage)$ to the nearest 10mA. 4) <i>SPR AVS</i> current for this voltage range is the maximum current as <i>Advertised</i> by the 15V <i>Fixed Supply</i> PDO. This current can be higher than 3A (refer to Note 2). Requires a 5A cable if over 3A is <i>Advertised</i> . 5) The <i>Sink</i> is allowed to request up to the 20V <i>Fixed Supply</i> Max Current when the requested voltage is 15.0V. 6) The Max Voltage for <i>SPR AVS</i> is what is allowed by Table 10.2, "SPR Normative Voltages and Minimum Currents" based on the port's Port Maximum PDP . 7) This <i>SPR Fixed Supply</i> voltage is only available if allowed by Table 10.2, "SPR Normative Voltages and Minimum Currents" based on the port's Port Maximum PDP . 8) <i>SPR Sources</i> May offer (A)PDOs at this Port Present PDP					

In reference to [Table 10.3, "SPR Source Capabilities When Port Present PDP is less than Port Maximum PDP"](#), [Table 10.4, "SPR Source Port Present PDP less than Port Maximum PDP Examples"](#) gives examples of which SPR capabilities are *Advertised* based on **Port Present PDP** on a *Managed Capability Port* and the port's **Port Maximum PDP** and cable's current rating.

Table 10.4 SPR Source Port Present PDP less than Port Maximum PDP Examples

Port Maximum PDP and Cable Rating	Port Present PDP	Offers				
		5V Fixed	9V Fixed	15V Fixed	20V Fixed	SPR AVS
80W / 5A	65W	3A ¹	3A ¹	3A ¹	3.25A	9V – 15V: 3A 15V – 20V: 3.25A
80W / 5A	40W	3A ¹	3A ¹	2.67A	2A	9V – 15V: 2.67A 15V – 20V: 2A
80W / 3A	40W	3A ¹	3A	2.67A	2A	9V – 15V: 2.67A 15V – 20V: 2A
40W / 5A	40W	3A ¹	3A ¹	2.67A	Not Offered	9V – 15V: 2.67A
40W / 3A	40W	3A ¹	3A	2.67A	Not Offered	9V – 15V: 2.67A
80W / 5A	20W	3A ¹	2.22A	1.33A	1A	9V – 15V: 1.33A 15V – 20V: 1A
80W / 3A	20W	3A ¹	2.22A	1.33A	1A	9V – 15V: 1.33A 15V – 20V: 1A
40W / 5A	20W	3A ¹	2.22A	1.33A	Not Offered	9V – 15V: 1.33A
40W / 3A	20W	3A ¹	2.22A	1.33A	Not Offered	9V – 15V: 1.33A
80W/3A	15W	3A	1.67A ²	1A ²	0.75A ²	9V - 15V: 1A ² 15V - 20V: 0.75A ²
40W/3A	15W	3A	1.67A ²	1A ²	Not offered	9V - 15V: 1A ²
1)	The <i>Fixed Supply</i> PDO Maximum Current field will <i>Advertise</i> at least 3A but <i>May Advertise</i> to RoundUp (PDP/voltage) to the nearest 10mA.					
2)	These <i>Capabilities</i> are not required but may be offered at this Port Present PDP .					

10.2.2.1 Fixed Supply PDOs

Figure 10.1, "SPR Source Power Rule Illustration for Fixed Supply PDOs" illustrates the minimum current that an SPR Source **shall** support at each voltage for a given PDP Rating for Fixed Supply PDOs.

Note: Not illustrated are that currents higher than 3A are allowed to be offered up to a limit of 5A given that a 5A cable is detected by the Source and the voltage times current remains within the Source PDP Rating.

Figure 10.1 SPR Source Power Rule Illustration for Fixed Supply PDOs

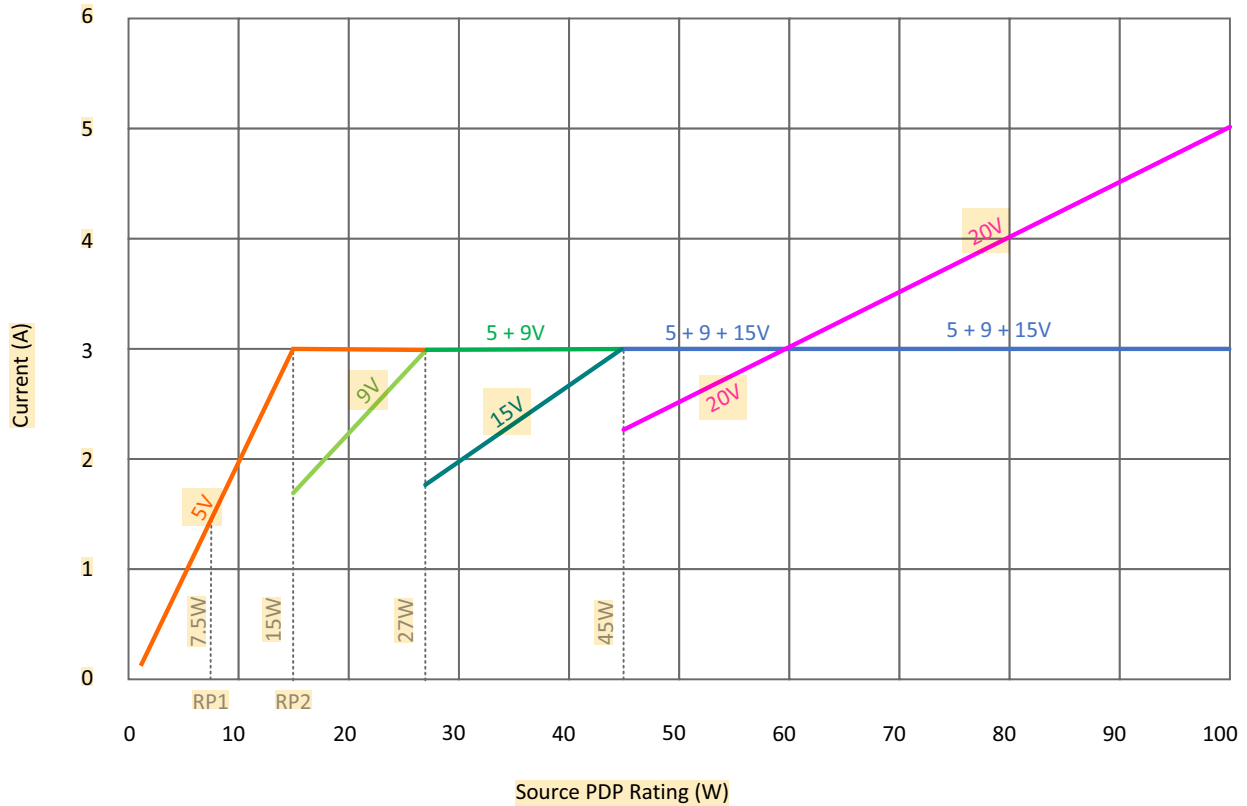


Figure 10.2, "SPR Source Power Rule Example For Fixed Supply PDOs" shows an example of an adapter with a rating at 50W. The adapter is required to support 20V at 2.5A, 15V at 3A, 9V at 3A and 5V at 3A.

Figure 10.2 SPR Source Power Rule Example For Fixed Supply PDOs

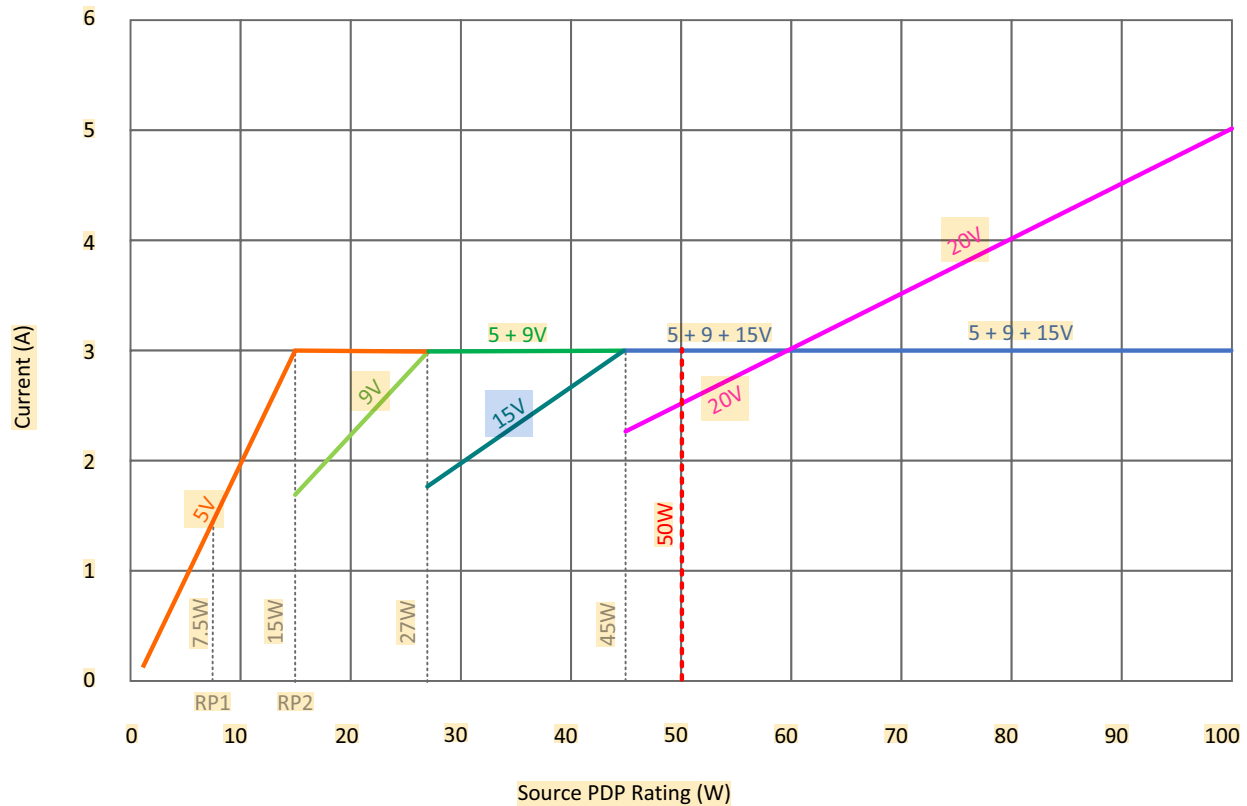


Table 10.5, "Fixed Supply PDO - Source 5V", Table 10.6, "Fixed Supply PDO - Source 9V", Table 10.7, "Fixed Supply PDO - Source 15V" and Table 10.8, "Fixed Supply PDO - Source 20V" show the Fixed Supply PDOs that **shall** be supported for each of the **Normative** voltages defined in Table 10.2, "SPR Normative Voltages and Minimum Currents".

Table 10.5 Fixed Supply PDO - Source 5V

Bit(s)	Description	
B31...30	<i>Fixed Supply</i>	
B29	<i>Dual-Role Power</i>	
B28	<i>USB Suspend Supported</i>	
B27	<i>Unconstrained Power</i>	
B26	<i>USB Communications Capable</i>	
B25	<i>Dual-Role Data</i>	
B24	<i>Unchunked Extended Messages Supported</i>	
B23	<i>EPR Capable</i>	
B22	<i>Reserved – Shall be set to zero.</i>	
B21...20	<i>Peak Current</i>	
B19...10	5V	
B9...0	Current based on PDP	
	PDP Rating (x)	Current (A)
	$0.5 \leq x \leq 15$	$x \div 5$
	$15 < x \leq 25$	$3 \leq A \leq x \div 5$
	$25 < x \leq 100$	$3 \leq A \leq 5$

Table 10.6 Fixed Supply PDO - Source 9V

Bit(s)	Description	
B31...30	<i>Fixed Supply</i>	
B29...22	Reserved – Shall be set to zero.	
B21...20	Peak Current	
B19...10	9V	
B9...0	Current based on <i>PDP</i>	
	PDP Rating (x)	Current (A)
	$0.5 \leq x \leq 15$	PDO not required
	$15 < x \leq 27$	$x \div 9$
	$27 < x \leq 45$	$3 \leq A \leq x \div 9$
	$45 < x \leq 100$	$3 \leq A \leq 5$

Table 10.7 Fixed Supply PDO - Source 15V

Bit(s)	Description	
B31...30	<i>Fixed Supply</i>	
B29...22	Reserved – Shall be set to zero.	
B21...20	Peak Current	
B19...10	15V	
B9...0	Current based on <i>PDP</i>	
	PDP Rating (x)	Current (A)
	$0.5 \leq x \leq 27$	PDO not required
	$27 < x \leq 45$	$x \div 15$
	$45 < x \leq 75$	$3 \leq A \leq x \div 15$
	$75 < x \leq 100$	$3 \leq A \leq 5$

Table 10.8 Fixed Supply PDO - Source 20V

Bit(s)	Description	
B31...30	<i>Fixed Supply</i>	
B29...22	Reserved – Shall be set to zero.	
B21...20	Peak Current	
B19...10	20V	
B9...0	Current based on <i>PDP</i>	
	PDP Rating (x)	Current (A)
	$0.5 \leq x \leq 45$	PDO not required
	$45 < x \leq 100$	$x \div 20$

More current **May** be offered in the PDOs when **Optional** voltages/currents are supported and a 5A cable is being used (see [Section 10.2.3, "Optional Voltages/Currents"](#)).

10.2.2.2

SPR Adjustable Voltage Supply (AVS)

For SPR AVS, [Figure 10.3](#), "Valid SPR AVS Operating Region for a Source advertising in the range of $27W < PDP \leq 45W$ ", [Figure 10.4](#), "Valid SPR AVS Operating Region for a Source advertising in the range of $45W < PDP \leq 60W$ " and [Figure 10.5](#), "Valid SPR AVS Operating Region for a Source advertising in the range of $60W < PDP \leq 100W$ " illustrate the valid operating region for SPR AVS RDO requests in the ranges of $27W < PDP \leq 45W$, $45W < PDP \leq 60W$ and $60W < PDP \leq 100W$, respectively.

Figure 10.3 Valid SPR AVS Operating Region for a Source advertising in the range of $27W < PDP \leq 45W$

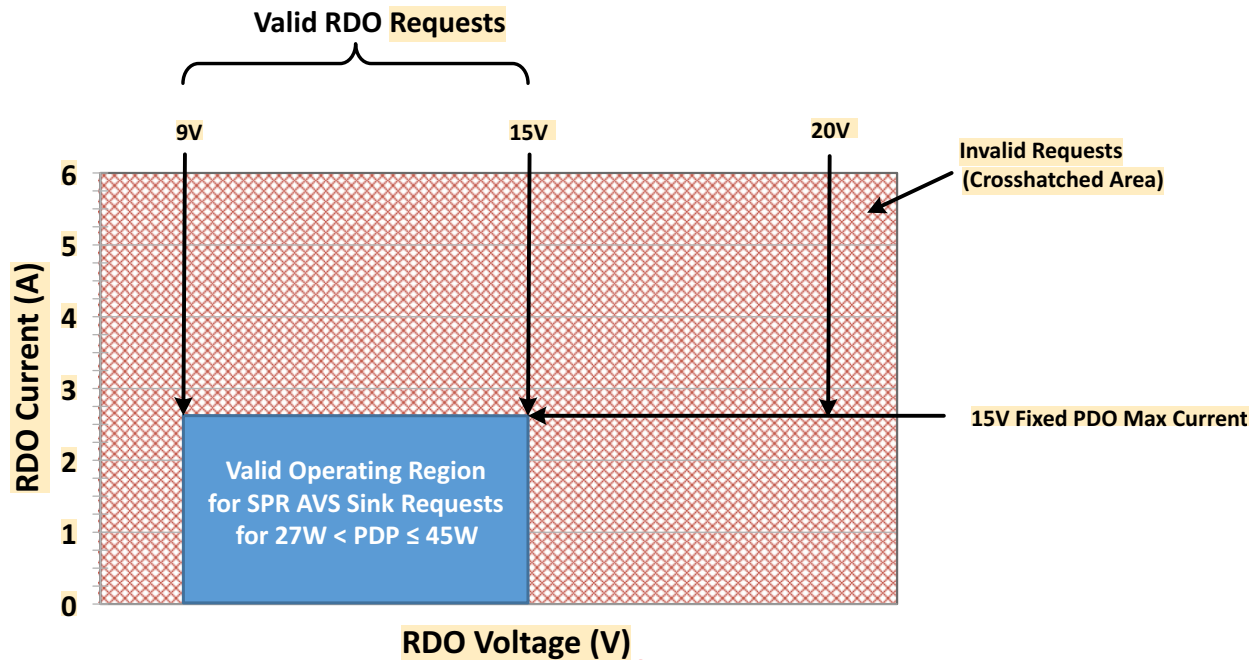


Figure 10.4 Valid SPR AVS Operating Region for a Source advertising in the range of $45W < PDP \leq 60W$

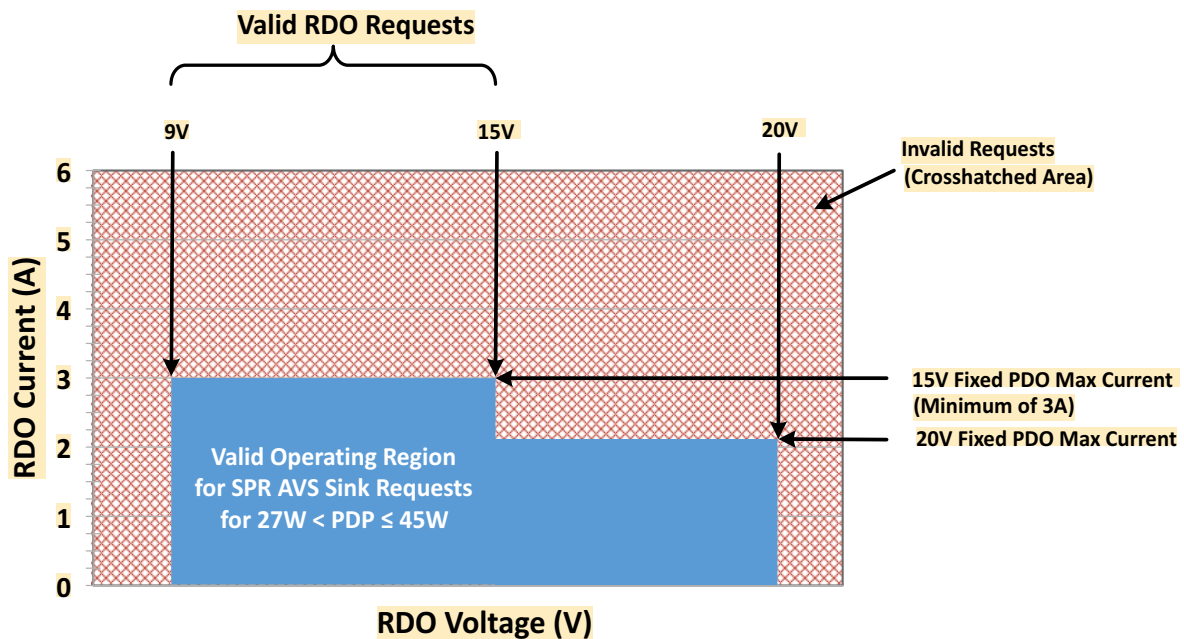
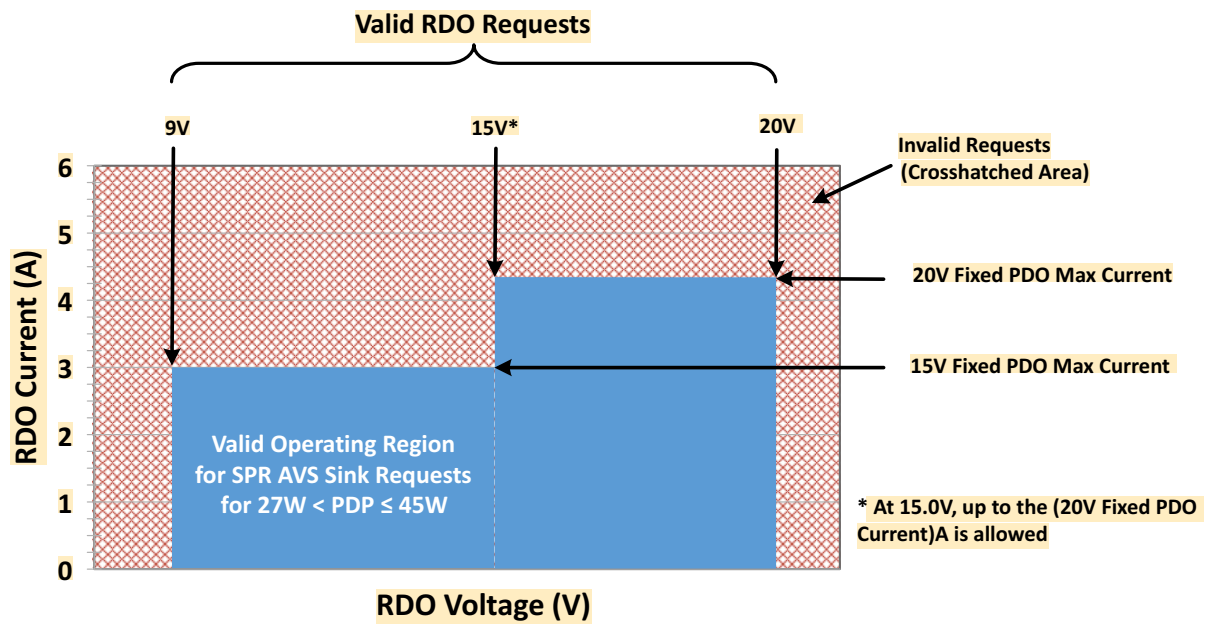


Figure 10.5 Valid SPR AVS Operating Region for a Source advertising in the range of $60W < PDP \leq 100W$



10.2.2.2.1 SPR Adjustable Voltage Supply (AVS) Voltage Ranges

Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable" shows the Minimum and Maximum Voltage for the SPR AVS ranges.

Table 10.9 SPR Adjustable Voltage Supply (AVS) Voltage Ranges

	AVS Voltage Range	
	15V AVS	20V AVS
Maximum Voltage	15V	20V
Minimum Voltage	9V	9V

The voltage output at the Source's connector **shall** be +/-5% for both the Maximum Voltage and the Minimum Voltage.

10.2.3 Optional Voltages/Currents

10.2.3.1 Optional Normative Fixed, Variable and Battery Supply

In addition to the **voltages** and currents specified in [Section 10.2.2, "Normative Voltages and Currents"](#), an **SPR Source** that is optimized for use with a specific **Sink** or a specific class of **Sinks** **May Optionally** supply additional **voltages** and increased currents. However, the **Optional** **voltages** **Shall Not** exceed 9V.

Optional **voltages** **Shall Not** be implemented on **EPR Source** including for both **SPR Mode** and **EPR Modes** of operation. **EPR** versions of **Variable Supply** and **Battery Supply** **PDOs** are not defined and **Shall Not** be implemented, however **SPR Variable Supply** and **Battery Supply** **PDOs** are allowed in **EPR Mode**.

While allowed, the use of **Optional** **voltages** and currents is not recommended as two **Sources** with the same **PDP Rating** but not supporting the same **Optional** **voltages** and currents can behave differently thus confusing the user.

See [Section 10.2, "Source Power Rules"](#) for the rules that **Shall** apply to **Optional** **PDOs** in order to be consistent with the declared **PDP Rating** and the **Normative** **voltages** and currents.

10.2.3.2 Optional Normative SPR Programmable Power Supply

The **voltages** and currents a Programmable Power Supply with a **PDP Rating** of x Watts **Shall** support are as defined [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#).

When **Optional** Programmable Power Supply **APDOs** are offered, the following requirements **Shall** apply:

- A **Source** that **Advertises** **Optional** Programmable Power Supply **APDOs** **Shall** **Advertise** the **PDOs** and **APDOs** shown in [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#).
- A **Source** **Shall** **Advertise** **Optional** Programmable Power Supply **APDOs** with Maximum Voltage and Minimum Voltages for nominal **voltage** as defined in [Table 10.11, "SPR Programmable Power Supply Voltage Ranges"](#).
- A **Source** **Shall Not** **Advertise** a Programmable Power Supply **APDO** that does not follow the Minimum Voltage and Maximum Voltage defined in [Table 10.11, "SPR Programmable Power Supply Voltage Ranges"](#).
- In no case **Shall** a **Source** **Advertise** a current that exceeds the **Attached cable's** current rating.
- The Max Voltage **Shall Not** exceed 21V while in **SPR Mode**.

Table 10.10 SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP

PDP Maximum PDP (W)	SPR Fixed and AVS	9V Prog ³	15V Prog ³	20V Prog ³
x < 15W	Required per <i>Table 10.2, "SPR Normative Voltages and Minimum Currents"</i> (or <i>Table 10.3, "SPR Source Capabilities When Port Present PDP is less than Port Maximum PDP"</i> when applicable)	-	-	-
15W		-	-	✖
15 < x < 27W		✖(PDP/9)A✖	-	-
27W		3A	-	-
27 < x < 45W		3A ²	(PDP/15)A ¹	-
45W		-	3A	-
45 < x < 60W		-	3A ²	(PDP/20)A ¹
60W		-	-	3A
60 < x < 100W		-	-	(PDP/20)A ¹
100W		-	-	5A
1) The <i>SPR PPS APDOs</i> Maximum Current field Shall Advertise RoundDown (PDP/Prog Voltage) to the nearest 50mA.				
2) The <i>SPR PPS APDOs</i> Maximum Current field Shall Advertise at least 3A, but May Advertise up to RoundDown(PDP/Prog Voltage) to the nearest 50mA.				
3) Applies to <i>APDOs</i> regardless of value of the PPS Power Limited bit.				

10.2.3.2.1 SPR Programmable Power Supply Voltage Ranges

The *SPR PPS* Voltage ranges map to the *Fixed Supply* Voltages. For each *fixed voltage* there is a defined *voltage range* for the matching *SPR PPS APDO*. [Table 10.11, "SPR Programmable Power Supply Voltage Ranges"](#) shows the Minimum and Maximum Voltage for the Programmable Power Supply that corresponds to the Fixed nominal voltage.

Table 10.11 SPR Programmable Power Supply Voltage Ranges

	Fixed Nominal Voltage		
	9V Prog	15V Prog	20V Prog
Maximum Voltage	11V	16V	21V
Minimum Voltage	5V	5V	5V

The *voltage* output at the *Source's* connector **shall** be +/-5% for both the Maximum Voltage and the Minimum Voltage.

10.2.3.2.2 Examples of the use of SPR Programmable Power Supplies

The following examples illustrate what a power adapter that *Advertises* a particular *PDP Rating* **May** offer:

- 1) *PDP 27W* implementation includes:
 - 5V @ 3A,
 - 9V @ 3A, and
 - 9V Prog @ 3A 
- 2) *PDP 36W* implementation includes:
 - 5V @ 3A,
 - 9V @ 3A,
 - 15 @ 2.4A,
 - *SPR AVS* with 9V - 15V @ 2.4A,
 - 9V Prog @ 3 A, and
 - 15V Prog @ 2.4A.
- 3) *PDP 36W* implementation that **Optionally** includes higher current in the 9V Prog PPS:
 - 5V @ 3A,
 - 9V @ 3A,
 - 15 @ 2.4A,
 - *SPR AVS* with 9V - 15V @ 2.4A,
 - 9V Prog @ >3A up to 4A (with a 5A cable) and 15V
 - Prog @ 2.4A.
- 4) *PDP 50W* implementation includes:
 - 5V @ 3A,
 - 9V @ 3A,
 - 15 @ 3A,
 - 20V @ 2.5A,

- *SPR AVS* with 9V - 15V @ 3A & 15V - 20V @ 2.5A,
- 15V Prog @ 3A, and
- 20V Prog @ 2.5A.

5) *PDP* 80W implementation includes:

- 5V @ 3A,
- 9V @ 3A,
- 15 @ 3A,
- 20V @ 4A,
- *SPR AVS* with 9V - 15V @ 3A & 15V - 20V @ 4A,
- 15V Prog @ 3A, and
- 20V Prog @ 4A.

The first example illustrates a basic example of a supply that can only support 5V and 9V.

The second and third examples illustrate as the *PDP Rating* goes higher there are more possible combinations that meet the *Power Rules*. These examples also add *SPR AVS*. Although there are multiple ways to meet the *Power Rules*, while operating in *SPR Mode* no more than a combination of seven *SPR (A)PDOs* and *APDOs* can be offered.

The fourth and fifth example show that the 15V Prog @ 3A fully covers the 9V Prog @ 3A range so it is not necessary to *Advertise* both. These examples also illustrate *SPR AVS* being extended up to 20V with separate current limits for the 9V - 15V and 15V - 20V ranges - a single *SPR AVS APDO* covers advertising both ranges.

10.2.3.3 Optional Normative Extended Power Range (EPR)

Support of *EPR Mode* is **Optional**. An *EPR Capable* port has a *PDP Rating* that is $>100\text{W}$ and $\leq 240\text{W}$. An *EPR Capable Source Port (EPR Source Port)* **May** operate in either *SPR Mode* or *EPR Mode* when operating at 100W or less.

An *EPR Source Port* operating in *SPR Mode* **May** offer less than 100W to avoid violating safety regulations. When operating in *EPR Mode*, an *EPR Source Port* **Shall** offer 100W in Fixed 20V when not constrained by multi- port sharing limits.

An *EPR Source* **May** include multiple ports and these ports can be functionally implemented as *Shared Capacity Charger* or *Assured Capacity Charger* ports as defined in [USB Type-C 2.4].

Any port on an *EPR Source* that has a *Port Present PDP* of 100W or less **Shall** follow the **Normative** requirements for *SPR Source Ports* and **Shall** operate only in *SPR Mode*. Any port on an *EPR Source* that is operating with a cable that is not *EPR Capable* **Shall** operate only in *SPR Mode*. An *EPR Source*, when operating in *SPR Mode* with a 5A cable, **May** offer less than 5A due to design tolerances in order to meet applicable safety standards. For best user experience it **Should** be as close to 100W as possible.

Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable" and Table 10.13, "EPR Source Capabilities when Port Present PDP is less than Port Maximum PDP and using an EPR-capable cable" define the **Normative** requirements for *EPR Source Ports*. While not included in these tables, any *EPR Source Port* that also supports *SPR PPS* **Shall** offer the *SPR Fixed 20V PDO* and *PPS 20V Prog APDO* at 100W (or the maximum available *PDP* when the port is operating at an Equivalent *PDP* $<100\text{W}$) when in *EPR Mode*:

- When an *EPR Source Port* is capable of supplying its *PDP Rating*, it **Shall** adhere to the requirements defined in Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable" based on its *PDP Rating* of x Watts.
- When a *Source Port* on an *EPR Charger* is unable to provide its *Port Maximum PDP*, it **Shall** adhere to the requirements defined in Table 10.13, "EPR Source Capabilities when Port Present PDP is less than Port Maximum PDP and using an EPR-capable cable" based on a *Port Present PDP* of x Watts. Some examples:
 - An *EPR Source Port* **May** be unable to provide its *PDP Rating* because it is thermally constrained at the time of power *Negotiation*.
 - A *Shared port* on a multi-port *EPR Charger* that is limited by the remaining available power.
- When an *EPR Charger* is in an *Adjustable Voltage Supply (AVS) Explicit Contract*:
 - It **Shall** Reject all Requests outside of the defined voltage range (see Table 10.15, "EPR Adjustable Voltage Supply (AVS) Voltage Ranges") or for a requested voltage and Current that results in a power level that is more than the Port's *Advertised PDP*.
 - In no case **Shall** a *Source* Advertise a Current or accept a Current requested by a *Sink* that exceeds the Attached cable's current rating.

- The Max Voltage offered by an *EPR Source* **Shall Not** exceed 48V

Table 10.12 EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable

Port Maximum PDP (W)	SPR Fixed and AVS	28V Fixed	36V Fixed ³	48V Fixed	EPR AVS ^{3, 4}
100 < x ≤ 140	Required per Table 10.2, "SPR Normative Voltages and Minimum Currents" (or Table 10.3, "SPR Source Capabilities When Port Present PDP is less than Port Maximum PDP" when applicable)	(PDP/28)A ²	N/A ¹	N/A ¹	(15V – PDP/5A): 5A (>PDP/5A – 28V): (PDP/AVS voltage) A
140 < x ≤ 180		5A	(PDP/36)A ²	N/A ¹	(15V – PDP/5A): 5A (>PDP/5A – 36V): (PDP/AVS voltage) A
180 < x ≤ 240		5A	5A	(PDP/48)A ²	(15V – PDP/5A): 5A (>PDP/5A – 48V): (PDP/AVS voltage) A
1) EPR Sources are disallowed from offering Fixed Supply voltages that are above the defined voltages for a given PDP, e.g., 36V is disallowed for any PDP of 140W or lower. 2) The Fixed PDOs Maximum Current field Shall Advertise either RoundDown (PDP/voltage) or RoundUp (PDP/voltage) to the nearest 10mA. 3) EPR Sources Shall reject any request for more than the Advertised PDP, i.e., when output voltage and operating current requested in the Sink RDO is outside of the defined AVS voltage and current range represented by the Advertised PDP, the RDO will be rejected. 4) The current available for a given AVS voltage is as indicated in this column. The current defined here is describing the top edge of the Valid Operating Region as illustrated in Figure 10.6, "Valid EPR AVS Operating Region". The AVS APDO does not have a Maximum Current field, so the maximum current has to be calculated from the PDP.					

Table 10.13 EPR Source Capabilities when Port Present PDP is less than Port Maximum PDP and using an EPR-capable cable

Port Present PDP (W)	SPR Fixed and AVS	28V Fixed	36V Fixed ⁴	48V Fixed ⁴	EPR AVS with Max Voltage of 28V, 36V or 48V per Table 10.12 ^{2, 5, 6}		
					28V	36V	48V ^x
7.5 ≤ x ≤ 15	Required per Table 10.2, "SPR Normative Voltages and Minimum Currents" (or Table 10.3, "SPR Source Capabilities When Port Present PDP is less than Port Maximum PDP" when applicable)	(PDP/28) A ³	(PDP/36) A ³	(PDP/48) A ³	(15V – max voltage): (PDP/AVS Voltage) A³		
15 < x ≤ 27							
27 < x ≤ 45							
45 < x ≤ 60		(PDP/28) A ¹					
60 < x ≤ 100			(PDP/36) A ¹				
100 < x ≤ 140	Table 10.3, "SPR Source Capabilities When Port Present PDP is less than Port Maximum PDP" with a Port Present PDP of 100W.	5A			Up to 75W: (15V – max voltage): (PDP/AVS voltage) A^x		
140 < x ≤ 180							
180 < x ≤ 240		5A	5A	(PDP/48) A ¹	Above 75W: (15V – PDP/5A): 5A (>PDP/5A – max voltage): (PDP/AVS voltage) A		

1) The Fixed Supply PDOs Maximum Current field **shall** Advertise either RoundDown (PDP/voltage) or RoundUp (PDP/voltage) to the nearest 10mA.

2) EPR SourceS **shall** reject any Request for more than the Advertised PDP, i.e., when output voltage and operating current requested in the Sink RDO is outside of the defined AVS voltage and current range represented by the Advertised PDP, the RDO will be rejected.

3) EPR SourceS **may** offer an (A)PDOs at this **Port Present PDP**. When offered, the Fixed Supply PDOs **Maximum Current** field **shall** Advertise either RoundDown (PDP/Voltage) or RoundUp (PDP/Voltage) to the nearest 10mA.

4) This EPR Fixed Supply voltage is only available if allowed by [Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable"](#) based on the port's PDP Rating.

5) The Max Voltage for AVS is what is allowed by [Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable"](#) based on the port's **Port Maximum PDP**.

6) The current available based on AVS voltage is as indicated in this column. The current defined here is describing the top edge of the **Valid** Operating Region as illustrated in [Figure 10.6, "Valid EPR AVS Operating Region"](#). AVS APDO does not have a Maximum Current field so the maximum current has to be calculated from the PDP.

Note: EPR Managed Capability Ports when power constrained are defined to offer higher voltages at lower **Port Present PDP** (as per [Table 10.13, "EPR Source Capabilities when Port Present PDP is less than Port Maximum PDP and using an EPR-capable cable"](#)) than the port's **Port Maximum PDP** (as per [Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable"](#)) because these voltages would otherwise be available if the **Managed Capability Port** power hadn't been constrained. **Managed Capability Ports** are required to be properly identified to the user based on the port's **Port Maximum PDP**.

In reference to [Table 10.13, "EPR Source Capabilities when Port Present PDP is less than Port Maximum PDP and using an EPR-capable cable"](#), [Table 10.14, "EPR Source Examples when Port Present PDP is less than Port Maximum PDP"](#) gives examples of which EPR Capabilities, in addition to the required SPR Fixed Supply PDOs and SPR AVS APDO, are Advertised based on **Port Present PDP** and the port's **Port Maximum PDP**.

Table 10.14 EPR Source Examples when Port Present PDP is less than Port Maximum PDP

Port Maximum PDP	Port Present PDP	Offers			
		28V Fixed	36V Fixed	48V Fixed	AVS
200W	108W	3.86A	3A	2.25A	48V@108W
160W	108W	3.86A	3A	Not offered	36V@108W
120W	108W	3.86A	Not offered	Not offered	28V@108W
200W	72W	2.57A	2A	1.5A	48V@72W
160W	72W	2.57A	2A	Not offered	36V@72W
120W	72W	2.57A	Not offered	Not offered	28V@72W
200W	36W	1.29A	1A ¹	0.75A ¹	48V@36W ¹

1) These Capabilities are not required but may be offered at this **Port Present PDP**.

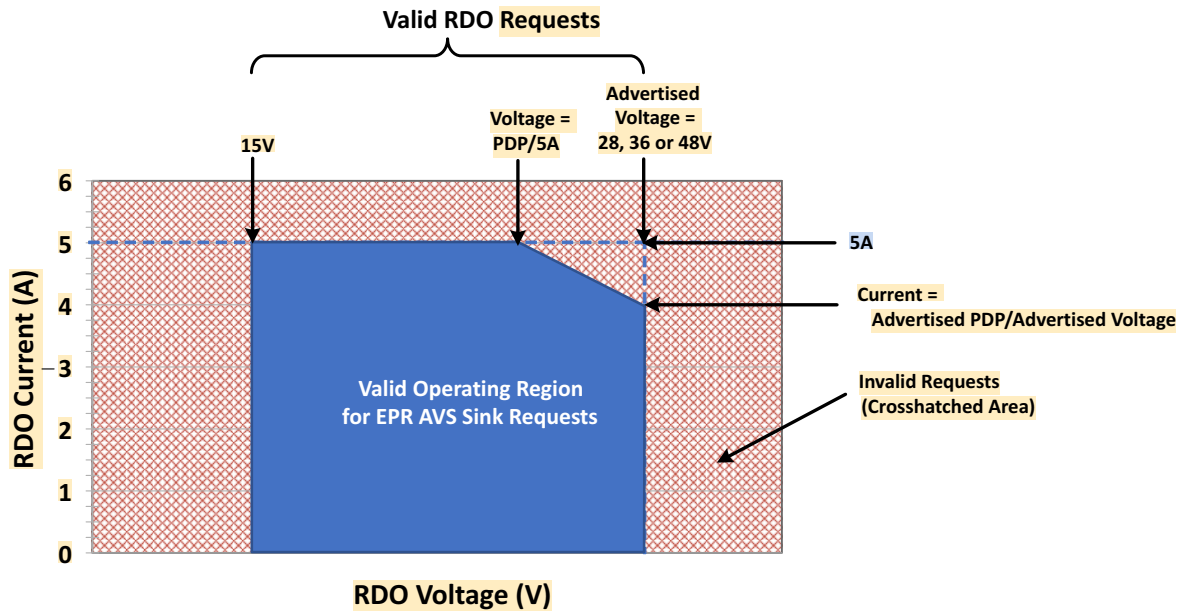
Table 10.14 EPR Source Examples when Port Present PDP is less than Port Maximum PDP

Port Maximum PDP	Port Present PDP	Offers			
		28V Fixed	36V Fixed	48V Fixed	AVS
160W	36W	1.29A	1A ¹	Not offered	36V@36W ¹
120W	36W	1.29A	Not offered	Not offered	28V@36W

1) These Capabilities are not required but may be offered at this **Port Present PDP**.

EPR Sources when operating in an AVS Explicit Contract are required to stay within their PDP as such they **Shall** respond to any request (VA) for more than the PDP with a **Reject** Message. [Figure 10.6, "Valid EPR AVS Operating Region"](#) illustrates the definition of the **Valid** operating range for an EPR Source operating in an AVS **Explicit Contract** based on its **Advertised PDP**.

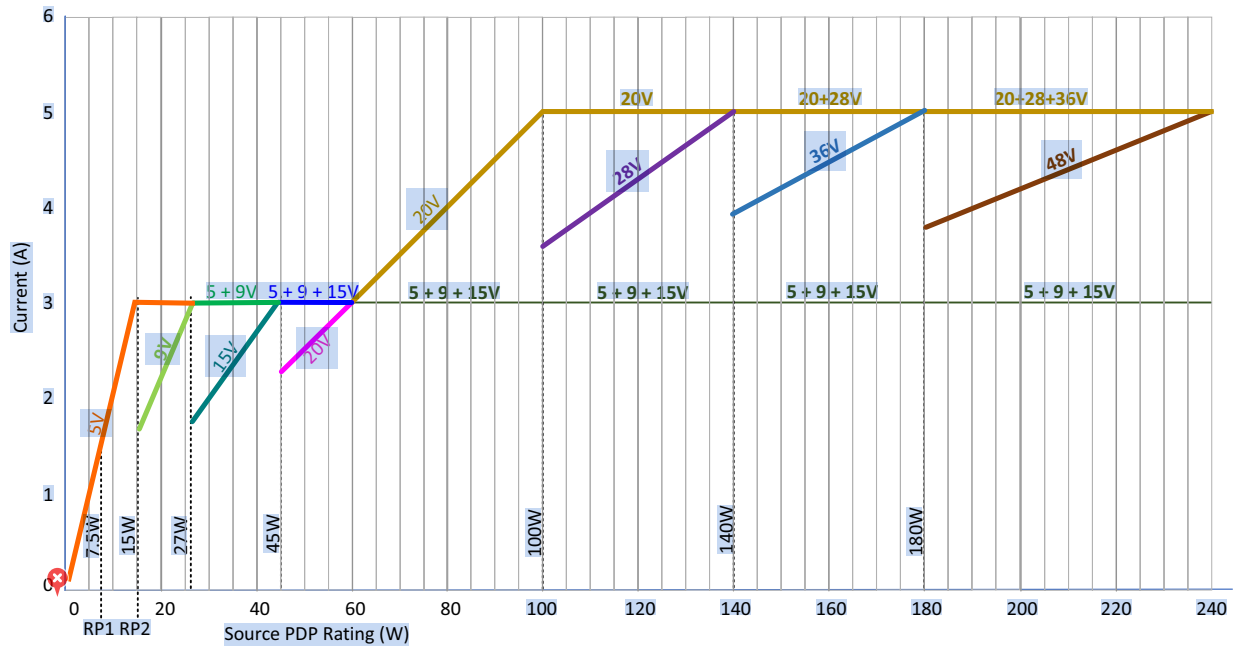
Figure 10.6 Valid EPR AVS Operating Region



[Figure 10.7, "EPR Source Power Rule Illustration for Fixed PDOs"](#) illustrates the minimum current that an EPR Source **Shall** support at each voltage for a given PDP Rating.

Note: Not illustrated are that currents higher than 3A are allowed to be offered up to a limit of 5A given that a 5A cable is detected by the Source and the voltage times current remains within the Source PDP Rating.

Figure 10.7 EPR Source Power Rule Illustration for Fixed PDOs



10.2.3.3.1 EPR Adjustable Voltage Supply (AVS) Voltage Ranges

Table 10.15, "EPR Adjustable Voltage Supply (AVS) Voltage Ranges" shows the Minimum and Maximum Voltage for the EPR AVS ranges.

Table 10.15 EPR Adjustable Voltage Supply (AVS) Voltage Ranges

	AVS Voltage Ranges		
	28V AVS	36V AVS	48V AVS
Maximum Voltage	28V	36V	48V
Minimum Voltage	15V	15V	15V

The voltage output at the Source's connector **Shall** be +/-5% for both the Maximum Voltage and the Minimum Voltage.

❗❗ 10.3 Sink Power Rules

10.3.1 Sink Power Rule Considerations

The *Sink Power Rules* are designed to ensure the best possible user experience when a given *Sink* used with a compliant *Source* of arbitrary Output Power Rating that only supplies the **Normative** voltages and currents.

The *Sink Power Rules* are based on the following considerations:

- Low power *Sources* (e.g., 5V) are expected to be very common and will be used with *Sinks* designed for a higher *PDP*.
- Optimizing the user experience when *Sources* with a higher *PDP Rating* are used with low power *Sinks*.
- Preventing *Sinks* that only function well (or at all) when using **Optional** voltages and currents.

10.3.2 Normative Sink Rules

Sinks designed to use *Sources* with a *PDP Rating* of x W **Shall**:

- Either operate or charge from *Sources* that have a *PDP Rating* $\geq x$ W.
- Either operate, charge or indicate a **Capabilities Mismatch** (see [Section 6.4.2.3, "Capability Mismatch"](#)) from *Sources* that have a *PDP Rating* $< x$ W and ≥ 0.5 W.

A *Sink* optimized for a *Source* with **Optional** voltages and currents or power as described in [Section 10.2.3, "Optional Voltages/Currents"](#) with a *PDP Rating* of x W **Shall** provide a similar user experience when powered from a *Source* with a *PDP Rating* of $\geq x$ W that supplies only the **Normative** voltages and currents as specified in [Section 10.2.2, "Normative Voltages and Currents"](#). For example, a 60W *Source* might not offer 9V Prog or 15V Prog since 20V Prog is a suitable substitute for both (as shown in [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#)).

The Operational Current/Power in the *Sink Capabilities* PDO for *Sinks* with an Operational *PDP* of x Watts **Shall** be as follows:

- Operational current for **Fixed Supply/Variable Supply** PDOs: $\text{RoundDown}(x/\text{voltage})$ to the nearest 10mA.
- Operational power for **Battery Supply** PDOs: $\leq x$.
- Operational current for Programmable Power Supply APDOs as defined in [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#): $\text{RoundDown}(x/\text{Prog Voltage})$ to the nearest 50mA.

The Maximum Current/Power in the *Sink RDO* for *Sinks* with an Operational *PDP* of x Watts and Maximum *PDP* of y Watts **Shall** be as follows:

- Maximum current for **Fixed Supply/Variable Supply** RDOs from *Sinks* without a *Battery*: $\text{RoundDown}(x/\text{voltage})$ to the nearest 10mA.
- Maximum current for **Fixed Supply/Variable Supply** RDOs from *Sinks* with a *Battery*: $\text{RoundDown}(y/\text{Voltage})$ to the nearest 10mA.
- Maximum power for **Battery Supply** RDOs from *Sinks* without a *Battery*: $\leq x$.
- Maximum power for **Battery Supply** RDOs from *Sinks* with a *Battery*: $\leq y$.
- Maximum current for PPS Supply RDOs from *Source* PDOs as defined in [Table 10.10, "SPR Programmable Power Supply PDOs and APDOs based on the Port Maximum PDP"](#) or [Table 10.14, "EPR Source Examples when Port Present PDP is less than Port Maximum PDP"](#): $\text{RoundDown}(y/\text{Prog Voltage})$ to the nearest 50mA❗

The following requirements **Shall** apply to the Advertised *Sink Capabilities*:

- A Sink **Shall Not** Advertise Fixed Supply PDO maximum voltages and currents that exceed the PDP Rating they were designed to use.
- A Sink **Shall Not** Advertise Variable Supply PDO maximum voltages and currents that exceed the PDP Rating they were designed to use.
- A Sink **Shall Not** Advertise a Battery Supply PDO maximum allowable power that exceeds the PDP Rating they were designed to use.
- A Sink **Shall Not** Advertise a PPS APDO maximum allowable power that exceeds the PDP Rating they were designed to use.
- A Sink **Shall Not** Advertise an AVS APDO maximum allowable power that exceeds the PDP Rating they were designed to use. ❗